Technical Specifications

CMPE 2965 - A02 - GROUP 8

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Project Technical Specification for a Noise Pollution Monitoring System

1. The status button will show the user the status of the battery, cellular connection, Bluetooth connection, and GNSS connection.
2. The device will use GNSS to receive accurate location and time data.
3. The device will measure the sound pressure in dB with a microphone once per second, and record the device location and current time when it takes a measurement.
4. The device will communicate with the server using a cellular connection to send each data packet.
5. The device will communicate via Bluetooth with the mobile app, which will be able to view the Cellular and GNSS status in more detail, as well as send date/time information to the device if GNSS time is not available.
6. The server will receive data from the device and sort it into recording sessions based on date, time, and location. Recording session data can be viewed and downloaded through the web app
7. The device will have a physical power control to turn it on/off, and a physical recording control to start/stop a recording session.
8. (Optional) The device will record and transmit audio to the server, which will attempt to classify possible sources of the sound, such as traffic, machinery, construction, or human activity.

Verification Test

Project : Noise Pollution Monitoring and Reporting Device
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1. The status button will show the user the status of the battery, cellular connection, Bluetooth connection, and GNSS connection.

Test	Push the status button. The cellular connection LED will show red or green to indicate the presence of a cellular connection, with green being connected and red being not connected. The Bluetooth LED will show red or blue to indicate a Bluetooth connection, with red being not connected and blue being connected. The GNSS LED will show red to indicate no connection, yellow to indicate poor connection quality (not very many GNSS satellites connected), and green to indicate a good connection. The four battery LEDs will light up to indicate battery charge level, with 1 LED indicating 0-25%, 2 indicating 25-50%, 3 indicating 50-75%, and 4 indicating 75-100%. The LEDs will be lit for 5 seconds.
Result	*To be determined*
Conclusion	*To be determined*

2. The device will use GNSS to receive accurate location and time data.

Test	Press the status button and confirm the GNSS status light is lit. Open the mobile control panel and read the location and time the device reports. Compare the location to a phone GPS reading taken at the same spot (within 10 m) and the time to a reference clock (within 1 second). Then confirm the data packet received by the server for that same timestamp contains the same location and time.
Result	*To be determined*
Conclusion	*To be determined*

3. The device will measure the sound pressure in dB with a microphone once per second, and record the device location and current time when it takes a measurement.

Test	Following the direct-measurement approach described in ISO 1996-2:2017, place the device beside a calibrated reference sound level meter and play a known steady tone (e.g. 70 dB) for 60 seconds. Record the device's logged dB values, timestamps, and location for each sample. Compare logged values against the reference meter, determine the measurement uncertainty per the standard's guidance, and confirm one reading is captured per second (±50 ms), with location and timestamp present on every entry.

Result	*To be determined*
Conclusion	*To be determined*

4. The device will communicate with the server using a cellular connection to send each data packet.

Test	With cellular connected (as determined by pushing the status button), run a 5-minute recording session. The server should receive one packet per second for the duration of the session.
Result	*To be determined*
Conclusion	*To be determined*

5. The device will communicate via Bluetooth with the mobile app, which will be able to view the Cellular and GNSS status in more detail, as well as send date/time information to the device if GNSS time is not available.

Test	The companion app, when connected over Bluetooth (as verified using the status button), displays cellular and GNSS connection information. If there is no GNSS connection, it can send location information from the phone it is running on to the device which is confirmed by looking at the packets sent to the server from that recording session.
Result	*To be determined*
Conclusion	*To be determined*

6. The server will receive data from the device and sort it into recording sessions based on date, time, and location. Recording session data can be viewed and downloaded through the web app.

Test	Run two recording sessions on different days/locations. On the web app, confirm each appears as a distinct, correctly grouped session, and that raw and interpreted data can be viewed and downloaded successfully for each.
Result	*To be determined*
Conclusion	*To be determined*

7. The device will have a physical power control to turn it on/off, and a physical recording control to start/stop a recording session.

Test	With the device off, activate the power control and confirm it boots (status LEDs activate for 5 seconds) within 3 seconds of the device being powered. With the device on and not recording, activate the recording switch and confirm a new session begins (a new session is logged on the server). Activate the recording control again and confirm the session ends cleanly (last sample logged, session marked complete on the server).
Result	*To be determined*
Conclusion	*To be determined*